

Maths of Cryptography

Caesar Cipher, Modular Arithmetic & RSA Public-Key Cryptography Preview

What is Cryptography? Cryptography is the mathematical science of protecting information by transforming readable text into secret codes, ensuring that only authorized recipients can read it. From ancient secret messages to modern digital banking and online security, cryptography keeps data safe!

1. Fundamental Terms in Cryptography

Term	Definition	Real-World / Simple Example
Plaintext	The original, unencrypted, readable message before transformation.	<i>CAT</i> or <i>HELLO PLANRS</i>
Encryption	The mathematical process of scrambling Plaintext into Ciphertext (<i>Plaintext</i> $\rightarrow$ <i>Ciphertext</i>).	Shifting each letter forward by a fixed key <i>k</i> .
Ciphertext	The encoded, scrambled message that appears unreadable to eavesdroppers.	<i>FDW</i> or <i>KHOOR SODQUV</i>
Decryption	The reverse process of converting Ciphertext back into readable Plaintext (<i>Ciphertext</i> $\rightarrow$ <i>Plaintext</i>).	Shifting letters backward using the secret key.

2. Caesar Cipher: Shift the Alphabet!

The Caesar Cipher is one of the oldest symmetric encryption techniques, named after Julius Caesar. It works by shifting every letter in the plaintext forward or backward by a fixed integer key *k* along the alphabet.

Example: Encrypting "CAT" with Shift Key $k = 3$

Let's shift each letter of **CAT** forward by +3 positions:

- **C** shifts 3 steps forward (*C* $\rightarrow$ *D* $\rightarrow$ *E* $\rightarrow$ *F*) $\rightarrow$ **F**
- **A** shifts 3 steps forward (*A* $\rightarrow$ *B* $\rightarrow$ *C* $\rightarrow$ *D*) $\rightarrow$ **D**
- **T** shifts 3 steps forward (*T* $\rightarrow$ *U* $\rightarrow$ *V* $\rightarrow$ *W*) $\rightarrow$ **W**

Plaintext: CAT $\rightarrow$ **Ciphertext:** FDW

3. Modular Arithmetic: Wrap-Around Like a Clock!

What happens when shifting a letter goes past Z? In cryptography, we use **Modular Arithmetic (Clock Arithmetic)**:

- Just like a 12-hour clock wraps around to 1 after 12 ($13 \equiv 1 \pmod{12}$), the 26-letter English alphabet wraps around to A after Z!
- **Example 1:** Shifting **Z** by +1 gives **A** (*Z* $\rightarrow$ *A*).
- **Example 2:** Shifting **Y** by +3 gives **B** (*Y* $\rightarrow$ *Z* $\rightarrow$ *A* $\rightarrow$ *B*).

Mathematical Formula: Assign numbers 0 to 25 to letters (A=0, B=1, ..., Z=25).

- Encryption: $C = (P + k) \bmod{26}$
- Decryption: $P = (C - k) \bmod{26}$

4. Why Caesar's Secret is Not So Secret!

Since there are only 26 letters in the English alphabet, there are only **25 possible non-zero shifts!**

- An eavesdropper or attacker can easily break a Caesar cipher by testing all 25 possible shifts in seconds (a *Brute-Force Attack*).
- Because of this weakness, modern communication requires asymmetric, computationally secure algorithms!

5. Preview of Modern Cryptography: RSA & The Magic Padlock

Modern encryption algorithms like **RSA** solve the security flaws of simple shift ciphers using **Asymmetric (Public-Key) Cryptography**. Instead of sharing one secret key, every user has a matching pair of keys:

Key Component	Public Key (The Open Padlock)	Private Key (The Secret Key)
Visibility / Sharing	Shared publicly with anyone who wants to send you a message.	Kept strictly secret by the owner; never shared with anyone.
Primary Function	Used exclusively to lock / encrypt messages.	Used exclusively to unlock / decrypt messages.
Padlock Analogy	Anyone can click the open padlock shut to secure a package.	Only the person holding the physical key can open the locked padlock.

The Padlock Analogy: Imagine Bob leaves an open padlock in a public space (**Public Key**). Alice takes Bob's padlock, puts her secret message in a box, and clicks the padlock shut. Now, nobody—not even Alice—can open the box without Bob's physical key. When Bob receives the box, he uses his unique private key (**Private Key**) to unlock it and read the message!

Cadet Challenge: Decode the Secret Message!

Apply your decoding skills to decrypt the ciphertext: **KHOOR SODQUV** using a backward shift of $k = 3$:

- $K(-3) \rightarrow H$, $H(-3) \rightarrow E$, $O(-3) \rightarrow L$, $O(-3) \rightarrow L$, $R(-3) \rightarrow O \rightarrow$ **HELLO**
- $S(-3) \rightarrow P$, $O(-3) \rightarrow L$, $D(-3) \rightarrow A$, $Q(-3) \rightarrow N$, $U(-3) \rightarrow R$, $V(-3) \rightarrow S \rightarrow$ **PLANRS**

Decoded Message: HELLO PLANRS

Key Takeaways:

- Cryptography protects data via Encryption (*Plaintext* → *Ciphertext*) and Decryption.
- Caesar Cipher shifts letters by a constant integer key k .
- Alphabet wrap-around is governed by modular arithmetic ($\pmod{26}$).
- Caesar ciphers are easily cracked because there are only 25 possible shift keys.
- RSA Public-Key Cryptography uses two distinct keys: a **Public Key** to lock/encrypt (shared with everyone) and a **Private Key** to unlock/decrypt (kept strictly secret).

Student Practice Worksheet

Question 1: Core Terminology

Define the terms *Plaintext*, *Ciphertext*, *Encryption*, and *Decryption* in your own words. How are these four terms related in a cryptographic pipeline?

Workspace / Student Answer:

Question 2: Caesar Cipher Encryption

Encrypt the following plaintext words using a Caesar cipher with a shift key of $k = 4$:

- a) **MATH** b) **SECRET**

Show step-by-step letter shifts:

Question 3: Alphabet Wrap-Around (Modular Arithmetic)

When encrypting letters near the end of the alphabet, wrap-around occurs. Determine the ciphertext or plaintext letter for each of the following:

- a) Encrypt letter **Z** with a shift key of $k = 2$.
b) Encrypt letter **Y** with a shift key of $k = 5$.
c) Decrypt letter **B** with a shift key of $k = 3$.

Workspace & Answers:

Question 4: Security Analysis & RSA Concepts

- a) Why is a Caesar cipher considered insecure for modern digital communication? How many total shift keys would an attacker need to test to guarantee breaking it?
- b) In RSA public-key cryptography, explain the exact roles of the **Public Key** and the **Private Key**. Which key is shared publicly to lock messages, and which key must be kept strictly secret to unlock them?

Workspace & Explanation: